

GOURI ANIL

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(225)650-5989

[Website](#)

EDUCATION

LOUISIANA STATE UNIVERSITY

PhD Candidate, Department of Oceanography and Coastal Sciences

CGPA: 4.18

Baton Rouge, LA

August 2022 - August 2026

Expected

COCHIN UNIVERSITY OF SCIENCE AND TECHNOLOGY

MSc in Physical Oceanography

CGPA: 9.69

Kerala, India

August 2019 - June 2021

MAR ATHANASIOS COLLEGE OF ARTS AND SCIENCES

BSc in Physics

CGPA: 9.43

Kerala, India

June 2016 - May 2019

RESEARCH EXPERIENCE

LOUISIANA STATE UNIVERSITY

PhD Candidate

Baton Rouge, LA

August 2022–Present

Assessing Impacts of Stratospheric Aerosol Injection (SAI) on Coral Bleaching and Marine Climate Exposure

Advisor: Dr. Cheryl S Harrison

Committee: Dr. Daniel Holstein, Dr. Junhong Liang, Dr. Gaurav Kandlikar

Project 1: Impacts of timing and level of SAI on Coral Thermal Bleaching

Project 2: Marine climate exposure under SAI and assessing socio-economic impacts to global nations

Project 3: Building an Observation-Based Model to Predict Coral Bleaching from Both Heat and Light Stress to Assess Impacts of SAI

INDIAN NATIONAL CENTRE FOR OCEAN INFORMATION SYSTEMS (INCOIS)

Graduate Researcher

Advisor: Dr. Abhisek Chatterjee (Senior Scientist, INCOIS)

Hyderabad, India

December 2020 – May 2021

Understanding Marine Heatwaves in the Arabian Sea

INDIAN INSTITUTE OF TECHNOLOGY

Undergraduate Summer Research Fellow

Advisor: Dr. Varun Bhalerao

Mumbai, India

May 2018 – July 2018

X-Ray data analysis of a pulsar Centaurus X-3 using NASA HEASoft

GRANTS AND FUNDING

2025 **Environmental Defense Fund**, Concept Lead and Lead Proposal Writer (Funded)

Preventing Coral Bleaching Through SAI: An Observation-Based Model to Predict Coral Bleaching from Both Heat and Light Stress to Assess Impacts of Stratospheric Aerosol Injection

PI: Dr. Cheryl S Harrison, Co-PI: Dr. Thomas Oliver

PUBLICATIONS

[Google Scholar](#)

Published

A. Chatterjee, **G. Anil**, L. R. Shenoy, *Marine Heatwaves in the Arabian Sea*, Ocean Science, 2022, DOI: 10.5194/os-18-639-2022

Accepted

K. E. Roberts, T. Rohr,... **G. Anil**, ..., *Potential impacts of climate interventions on marine ecosystems*. ESS Open Archive, November 25, 2025. Accepted for publication in *Reviews of Geophysics*. DOI: 10.22541/essoar.176401886.60047293/v1

Submitted

G. Anil, C.S. Harrison, J.A. Kleypas, D. Holstein, T.A. Oliver, V. Garza, K.E. Roberts, T. Felgenhauer, D. Jayewardene, *Impacts of Timing and Level of Stratospheric Aerosol Injection on Coral Thermal Bleaching*, ESS Open Archive. December 17, 2025. DOI: 10.22541/essoar.176599087.79155141/v1

L. Kounta, L. Luo, **G. Anil**, D. Hueholt, C. S. Harrison, D. Visioni, M. Tye, T. Felgenhauer, A. T. Gaye, P.L. Zarnetske, *Climate Intervention through Stratospheric Aerosol Injection may partially mitigate marine heatwaves*, ESS Open Archive. November 29, 2025. DOI: 10.22541/essoar.176442822.27571009/v1

PRESENTATIONS

ORAL

2025 AGU Fall Meeting
LSU 3-Minute Thesis Competition Finals
LSU College of Coast and Environment Annual Graduate Student Symposium
2024 Reef Futures Symposium

POSTERS

2025 AGU Fall Meeting
2024 LSU Graduate Research Conference
2024 AGU Ocean Sciences Meeting

MEDIA COVERAGE ON GRADUATE RESEARCH

2025 Could Stratospheric Aerosol Injection Help Save Corals from Bleaching? Eos. [Article](#)
2023 Marine heat waves may be favouring more cyclones in the Arabian Sea: INCOIS scientists. *The Hindu*. [Article](#)
2022 'At least 20 marine heatwave days in Arabian Sea during last decade.' *The Times of India*. [Article](#)
2022 The Arabian Sea faces an increased frequency and duration of marine heatwave days, finds study. *Mongabay-India*. [Article](#)

AWARDS AND FELLOWSHIPS

2025 Travel Award to attend the 16th Geoengineering Model Intercomparison Project Meeting, Tokyo
2025 NCAR High-Performance Computing Large Allocation for Climate Intervention Simulations
2025 Travel Award to attend the 2025 Degrees Global Forum, South Africa
2024 Graduate Student Organization Travel Award
2023 Travel Award to attend the Scripps Marine Geoengineering Workshop
2022 Graduate Excellence Award for Academic Achievement
2019 Indian Academy of Sciences Summer Research Fellowship
2018 Indian Academy of Sciences Summer Research Fellowship

PEER REVIEW

2025 Co-reviewer for Journal of Geophysical Research – Oceans
2025 Reviewer for American Academy of Arts & Sciences Solar Geoengineering Primer

TEACHING AND MENTORING

- 2026 Teaching Assistant and Guest Lecturer, Gulf of Mexico (Spring Semester)
- 2025 Undergraduate Honors Thesis Mentor (December 2025 to April 2026)
- 2024 Teaching Assistant, Introduction to Oceanography (Fall Semester)
- 2023 Teaching Assistant, Introduction to Oceanography (Fall Semester)

WORKSHOPS

- 2024 Community Climate Intervention Strategies (CCIS) Workshop, Notetaker
- 2023 Scripps Marine Geoengineering Scenario Design Workshop, Rapporteur
- 2019 International Conference on Frontiers in Marine Science, Volunteer and Session Host

OUTREACH

- 2024 Student Host for Ocean Commotion Outreach Event by LSU
- 2024 Student Host for LSU College of Coast and Environment Open House
- 2024 Coast and Environment Graduate Organization Secretary

TECHNICAL SKILLS

Programming: Python, R

Modeling: Experience running Community Earth System Model Simulations, Model Data Processing

Climate Data Processing and Visualization: netCDF, xarray, geopandas, matplotlib, CDO

Spatial Tools: QGIS